

Date

Note

18. 求曲线 $y = x^{\frac{2}{3}}$ 在点 $(1, 1)$ 处的曲率圆方程

解. $y' = \frac{2}{3}x^{-\frac{1}{3}}$, $y'' = -\frac{2}{9}x^{-\frac{4}{3}}$

在点 $(1, 1)$ 处有

$$y'|_{x=1} = \frac{2}{3}, \quad y''|_{x=1} = -\frac{2}{9}$$

曲率为

$$k = \frac{|y''|}{[1+(y')^2]^{\frac{3}{2}}} \Big|_{x=1} = \frac{|-\frac{2}{9}|}{[1+(\frac{2}{3})^2]^{\frac{3}{2}}} = \frac{6}{13\sqrt{13}}$$

曲率半径为

$$R = \frac{1}{k} = \frac{13\sqrt{13}}{6}$$

曲率中心坐标为

$$\alpha = \left\{ x - \frac{y'[1+(y')^2]}{y''} \right\} \Big|_{x=1} = 1 - \frac{\frac{2}{3}[1+(\frac{2}{3})^2]}{-\frac{2}{9}} = \frac{16}{3}$$

$$\beta = \left\{ y + \frac{1+(y')^2}{y''} \right\} \Big|_{x=1} = 1 + \frac{1+(\frac{2}{3})^2}{-\frac{2}{9}} = -\frac{11}{2}$$

故曲率圆方程为

$$(x - \frac{16}{3})^2 + (y + \frac{11}{2})^2 = \frac{2197}{36}$$